

1. Add a heading at the top:
Lab 4 – Vibe Coding Chat History
2. Make sure the conversation includes **all 4 tasks**:
 - Task 1 – Bézier path fix
 - Task 2 – Enemy tint
 - Task 3 – Wave start
 - Task 4 – Shield charges
3. Export/download it as **PDF**.
4. Name it:
Chat-History.pdf
5. Upload that PDF into your GitHub:
SE-PES1UG24CS638 → Lab-4

Then your Lab-4 folder will have all three required deliverables:

-  Before video
-  After video
-  Updated game.py
-  Chat-History.pdf

Actually that's **four files**, but they cover the three categories in the instructions.

If you want, after you create the PDF, **send it here before uploading** and I can check whether it contains everything needed.

HEyy give me one summarized conversation please like 4 to 5pages

Yesss 👍 I can give you a **clean 4–5 page Chat History document** that you can copy into Word/Google Docs and export as PDF.

Use this as your **Chat-History.pdf** content.

LAB 4 – VIBE CODING

Chat History / Prompt and Implementation Summary

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Course: Software Engineering

Lab: Lab 4 – Vibe Coding

Repository: 31_galaga

Vibe Coding Tool: ChatGPT

1. Introduction

Student Prompt

I have been assigned Lab 4 – Vibe Coding. The objective is to use a Vibe Coding tool such as ChatGPT to understand, fix, and improve a Python-based game.

The assigned project is a **Galaga-like Pygame game**. The repository contains one deliberate bug and three empty functions that need to be implemented.

The requirements are:

1. Clone or fork the assigned repository.
2. Run the Python game before making changes.
3. Record a 10-second video before modifications.
4. Fix the broken code using prompts.
5. Implement all the features mentioned in the README.
6. Record another video after completing the modifications.
7. Push the updated code to the personal GitHub repository.
8. Make separate commits for each task.
9. Do not create a pull request to the original SETAPESU26 repository.

ChatGPT Response / Guidance

The first step was to fork the assigned repository into the student's personal GitHub account.

The assigned repository was:

SETAPESU26/31_galaga

The personal fork created was:

tejaswini2006-hub/31_galaga

The repository was cloned locally and opened in Visual Studio Code.

The Python environment was checked and Pygame was installed. The command:

`python game.py`

was used to run the game successfully.

The game was tested before making any changes, and a before-change video was recorded.

2. Task 1 – Fixing the Bézier Curve Path

Problem Identified

The README stated that there was a bug in the enemy entry/dive path.

The game uses a **cubic Bézier curve** to control the movement of enemies.

The standard cubic Bézier formula is:

$$\begin{aligned} B(t) = & (1-t)^3 P_0 \\ & + 3(1-t)^2 t P_1 \\ & + 3(1-t)t^2 P_2 \\ & + t^3 P_3 \end{aligned}$$

In the original code, the third term was incorrect.

The original implementation contained:

```
3 * u * t * p2[0]
```

and

```
3 * u * t * p2[1]
```

The term was missing one multiplication by t.

Prompt Used

Please inspect the Bézier curve calculation in the game and fix the entry/dive path bug according to the standard cubic Bézier formula. Do not modify unrelated parts of the code.

Solution

The incorrect third term was changed to:

```
x = u ** 3 * p0[0] + 3 * u * u * t * p1[0] + 3 * u * t * t * p2[0] + t ** 3 * p3[0]
```

```
y = u ** 3 * p0[1] + 3 * u * u * t * p1[1] + 3 * u * t * t * p2[1] + t ** 3 * p3[1]
```

The important correction was:

```
3 * u * t
```

to:

```
3 * u * t * t
```

Testing

The game was executed again using:

```
python game.py
```

The game opened normally and the enemy movement worked correctly.

Git Commit

A separate commit was created:

Fix Bezier curve path calculation

The changes were pushed to the personal GitHub repository.

3. Task 2 – Implementing Enemy Tint

Problem Identified

The README contained an empty function:

```
def enemy_tint(kind):  
    pass
```

The function is called when an enemy is drawn.

The requirement was to return an RGB colour override depending on the enemy type.

The available enemy types were:

boss

red

blue

Prompt Used

Implement the `enemy_tint(kind)` function. It should return an RGB colour override for boss, red, and blue enemies, and return `None` for unknown types. Keep the existing game structure unchanged.

Solution

The function was implemented as:

```
def enemy_tint(kind):
```

```
if kind == "boss":  
    return (255, 215, 0)  
elif kind == "red":  
    return (255, 100, 100)  
elif kind == "blue":  
    return (100, 180, 255)  
return None
```

This produces:

Enemy	Colour
Boss	Gold
Red enemy	Light red
Blue enemy	Light blue

The existing drawing code already contained:

```
color = enemy_tint(enemy.kind) or ENEMY_COLORS[enemy.kind]
```

Therefore, the new function directly affected the enemy appearance without changing the rest of the game.

Testing

The game was run again.

The enemies appeared with the newly implemented colours and the game continued to function normally.

Git Commit

A separate commit was created:

Implement enemy tint colors

The changes were pushed to the personal repository.

4. Task 3 – Implementing Wave Start Notification

Problem Identified

The README contained another empty function:

```
def on_wave_start(wave):  
    pass
```

This function is called whenever a new wave begins.

The function can be used for things such as:

- Displaying a wave banner
- Increasing difficulty
- Changing colours
- Giving information about the new wave

For this assignment, a simple wave-start notification was implemented.

Prompt Used

Implement `on_wave_start(wave)` so that it gives a clear indication whenever a new wave starts. Keep the implementation simple and do not change unrelated game logic.

Solution

The function was changed to:

```
def on_wave_start(wave):  
    print(f"Wave {wave} started!")
```

This prints a message in the terminal whenever a wave starts.

For example:

Wave 1 started!

When the player completes a wave, the next wave is spawned and the function is called again.

Testing

The game was executed.

The terminal displayed:

Wave 1 started!

The game itself continued running normally.

This confirmed that the function was being called when the wave was created.

Git Commit

A separate commit was created:

Add wave start notification

The changes were pushed to the personal repository.

5. Task 4 – Implementing Shield Charges

Problem Identified

The final empty function was:

```
def shield_charges(wave):  
    pass
```

The README specified that the function should determine how many hits the player's shield can absorb during each wave.

The existing game already contained shield handling inside:

```
hit_player()
```

Therefore, only the number of shield charges needed to be implemented.

Prompt Used

Implement `shield_charges(wave)` so that the player receives one shield charge every three waves. Return 0 on other waves. Keep the existing shield consumption logic unchanged.

Solution

The function was implemented as:

```
def shield_charges(wave):  
    return 1 if wave % 3 == 0 else 0
```

This means:

Wave Shield Charges

1	0
2	0
3	1
4	0
5	0

Wave Shield Charges

6	1
7	0
8	0
9	1

Therefore, the player receives a shield on every third wave.

The existing code already resets the shield when a new wave begins:

```
self.shield = shield_charges(self.wave) or 0
```

Testing

The game was executed after implementing the function.

The game opened normally and the existing gameplay mechanics continued to work.

Git Commit

A separate commit was created:

Add shield charges by wave

The changes were pushed to the personal GitHub repository.

6. Final Testing and GitHub Submission

After completing all four tasks, the complete game was tested again.

The following features were successfully implemented:

Task 1

Fixed cubic Bézier path calculation

The enemy movement path now uses the correct cubic Bézier equation.

Task 2

Implemented enemy tint

Enemies receive different RGB colour overrides based on their type.

Task 3

Implemented wave start notification

The game prints a message whenever a new wave starts.

Task 4

Implemented shield charges

The player receives one shield charge every third wave.

Git Commit History

Separate commits were made for each task as required.

Fix Bezier curve path calculation

Implement enemy tint colors

Add wave start notification

Add shield charges by wave

All changes were pushed to the student's personal fork:

https://github.com/tejaswini2006-hub/31_galaga

No pull request was created to the original SETAPESU26 repository.

7. Before and After Demonstration

Before making any modifications, the original game was executed successfully.

A video was recorded to demonstrate the original game.

After implementing all four tasks, the modified game was executed again and another video was recorded.

The two videos demonstrate the state of the game:

Before → Original game

After → Modified game with implemented features

The videos were uploaded to the student's Software Engineering lab repository under the Lab-4 folder.

8. Final Deliverables

The Lab-4 folder contains the required submission materials:

Lab-4/

|
├— before.mp4.mp4
├— after.mp4.mp4
├— game.py
├— Chat-History.pdf
└— README.md

The updated game.py contains all four completed tasks.

The before and after videos demonstrate the game before and after modification.

The Chat History PDF documents the prompts, explanations, implementation steps, testing, and GitHub workflow used during the Vibe Coding assignment.

Conclusion

The Lab 4 Vibe Coding assignment was completed using ChatGPT as the Vibe Coding tool.

The project involved understanding an existing Pygame-based Galaga game, identifying the broken Bézier curve calculation, implementing three missing functions, testing the modified game, creating separate Git commits, and pushing the final code to a personal GitHub repository.

The four implemented improvements were:

1. **Correct cubic Bézier enemy movement**
2. **Enemy colour tinting**
3. **Wave start notification**
4. **Wave-based shield charges**

The final project was successfully tested and uploaded to the personal GitHub repository without creating a pull request to the original repository.